java Programming Project Presentation

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- Project Title: PERSONAL CONTACT DIRECTORY WITH SEARCH AND UPDATE
- Course:java Programming Lab / Mini Project
- Student Name & Roll No: NIHA 25BCS242
- Department: Computer Science
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Introduction

This project focuses on creating a simple contact directory using the C programming language. It allows users to store, search, update, and manage personal contact information efficiently.

The system replaces traditional manual contact storage with a digital solution, making retrieval faster and more organized.

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Objective

- To develop a C-based contact management system
- To store and manage personal contact details
- To implement search and update features
- To practice file handling and structured programming

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Problem Statement

Managing contacts manually or without organization can lead to:

- Difficulty in searching contacts quickly
- Data loss or duplication
- Lack of update flexibility

This project solves these issues by providing a structured and searchable contact directory.

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System Requirements

Hardware Requirements:

- Processor: Intel i3 or above
- RAM: 4 GB minimum
- Storage: 500 MB free space

Software Requirements:

- Operating System: Windows / Linux
- Compiler: GCC / Turbo C / Code::Blocks

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System Design

The system works in the following flow:

- User selects an option (Add, Search, Update, Delete, View)
- Program processes the request
- Data is stored/retrieved from file

Modules:

Input Module

Processing Module

Output Module

File Handling Module

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Module Description

- Input Module:

Takes contact details like name, phone number, email

- Processing Module:

Performs operations like search, update, delete

- Output Module:

Displays contact information

- File Handling Module:

Stores and retrieves contacts using files

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Algorithm / Flowchart

- Start
- Display menu
- User selects option
- Perform selected operation:
  - 1.Add contact
  - 2.Search contact
  - 3.Update contact
  - 4.Delete contact
  - 5.View all contacts
- Save changes to file
- Repeat until exit
- Stop

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Implementation

Concepts used:

- Structures (struct)
- Arrays
- Functions
- File handling (fopen, fprintf, fscanf, fclose)
- Conditional statements and loops

Example features:

- Add new contact
- Search by name or phone number
- Update existing contact details
- Delete contact

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Output Screens

Menu screen

Add contact screen

Search result screen

Updated contact display

File storage proof

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Results & Discussion

- Contacts stored successfully in file
- Fast search using name matching
- Update feature works efficiently
- Easy to use and user-friendly interface

Challenges faced:

- File handling errors
- String comparison issues

Solutions:

- Proper use of file modes
- Using strcmp() for comparison

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Conclusion & Future Enhancement

Conclusion:

- Successfully implemented a contact directory
- Improved understanding of file handling and structures
- Achieved project objectives

Future Enhancements:

- Add graphical user interface (GUI)
- Store data in database (MySQL)
- Add password protection
- Implement sorting and filtering options

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References / Acknowledgement

References:

- Let Us C – Yashavant Kanetkar
- C Programming – E. Balagurusamy
- Online tutorials and documentation

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Thank You